

SAPIENZA UNIVERSITY OF ROME | COMPUTER VISION

LIGHTWEIGHT DIFFUSION MODELS

Accelerating Training & Inference for Resource-Constrained Environments

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Code Repository: github.com/anilegin/latent-diffusion

Standard DDPMs are computationally expensive and slow

THE COMPUTATIONAL BOTTLENECK

- **Iterative Sampling:** Standard DDPMs require **1,000 sequential denoising steps** for a single image, making real-time inference unviable.
- **Huge Computational Footprint:** Standard architectures like Stable Diffusion use **850M+ parameter U-Nets**, requiring high-end GPUs.
- **Edge Obstacles:** Large VRAM requirements (>6GB) prevent localized running on consumer computers and edge devices.
- **Slow Training Cycle:** Training standard diffusion models from scratch takes weeks and massive compute power.

RESEARCH GOALS & DATASET

- **Architectural Novelty:** Build and train a compact **latent diffusion text-to-image pipeline** from scratch.
- **Compress Dimensions:** Downsample input images into a lower-dimensional latent space to accelerate training and sampling.
- **Ablation Study:** Conduct a rigorous evaluation of sampling acceleration (DDIM) and step reductions (50 to 150 steps).
- **Target Dataset:** Trained on **COCO 2017 captions**, featuring ~118,000 training images and 5,000 validation images resized to 256×256.

SOTA Text-to-Image Models

Frontier Text-to-Image Landscape (2026)

- **Black Forest Labs (FLUX.1 / FLUX.2):** Unmatched open-weights standard for photorealism and typographic spelling accuracy.
- **Midjourney v8.1 / v7:** The gold standard for high-fidelity aesthetics, artistic style control, and atmospheric lighting.
- **OpenAI GPT Image 2 / DALL-E 3:** Leading prompt alignment, robust multi-subject instruction following, and logical layout composition.
- **Google DeepMind Imagen 4:** State-of-the-art natural textures, fine product rendering, and photorealistic spatial coherence.

How SOTA Models Create Images

- **Diffusion Transformers (DiT):** Replaces convolutional UNets with self/cross-attention blocks. Images are treated as arrays of visual token patches, allowing models to scale predictably like LLMs.
- **Flow Matching / Rectified Flow:** Replaces stochastic SDE denoising paths with straight, deterministic vector fields. By training the network to predict straight lines, sampling steps are cut to 4-10 steps.
- **Multimodal Joint Attention (MMDiT):** Maintains separate parameter paths for language embeddings and image latents, resulting in a dramatic leap in visual spelling and prompt alignment.

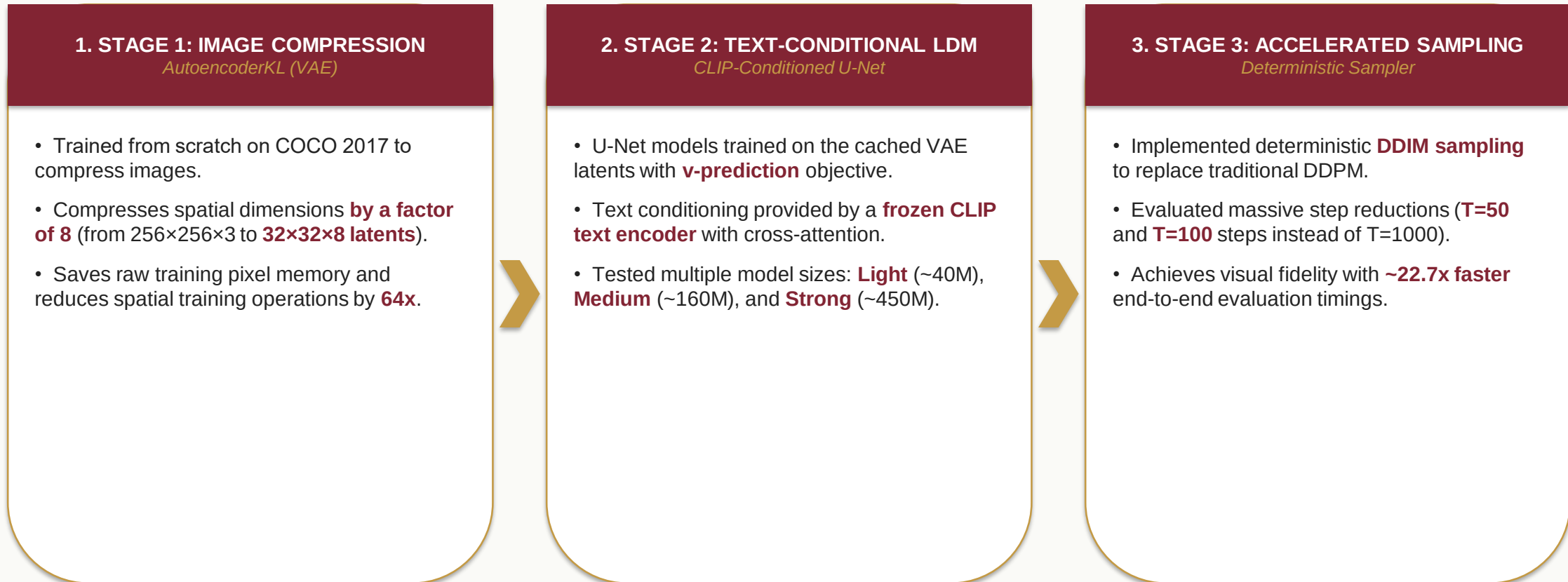
Visual Demonstration & Prompt Verification



Benchmark Prompt: "A hyper-detailed glass sphere resting on wet pavement reflecting neon-lit city signs, showcasing extreme photorealism, text-in-image rendering, and complex path-traced lighting reflections."

Analysis: Flow Matching and DiT models represent a monumental shift over classic U-Nets. They synthesize crisp text tokens (like 'SOTA 2026') without structural gibberish, while attention layers map physical raytracing vectors across complex surfaces.

A three-stage pipeline compresses images 8x and accelerates sampling ~22x



COCO 2017 Dataset Provides Rich Text-Guided Everyday Context

DATASET DESIGN & SPECIFICATIONS

- What is COCO?: Common Objects in Context is a premier vision dataset featuring everyday scenes with complex annotations.
- Training Scale: Standardized 2017 split containing 118,000 training images with multi-caption sentences.
- Validation Pool: Includes 5,000 distinct validation images used for evaluating FID and PSNR scores.
- Formatting constraints: Raw images are rescaled and center-cropped to 256x256 before latent caching.



A young boy standing in front of a computer keyboard



A red helmet is on a yellow toilet in the dirt



A silver and red bus in parking lot next to cars.



Two giraffe standing next to each other on a grassy field

Custom 8-channel VAE achieves 8.91 rFID with a 1.032 scaling factor

VAE ARCHITECTURE

- **Training Setup:** Trained completely **from scratch** on COCO 2017 training images for **~24 hours on a single A100 GPU**.
- **High Latent Capacity:** Features **8 latent channels** (instead of standard 4) to capture richer spatial structures before diffusion.
- **Latent Scaling Factor:** Estimated scaling factor of **1.032** to normalize latents to unit variance, ensuring stable noise scheduler work.
- **Evaluation:** Selected the best checkpoint based on validation reconstruction quality over 5,000 COCO images resized to 256×256.

RECONSTRUCTION PERFORMANCE VS PUBLIC BASELINES

Model	Params	rFID ↓	PSNR ↑	SSIM ↑	Perceptual ↓
Custom COCO VAE	98.19M	8.91	30.1 ± 3.5	0.88 ± 0.07	LPIPS 0.13 ± 0.04
LDM text2image VQ-VAE	83.65M	4.51	25.55 ± 3.62	0.757 ± 0.114	LPIPS 0.143 ± 0.042
SD VAE Original	83.65M	4.99	23.40 ± 3.80	0.690 ± 0.140	PSIM 1.01 ± 0.28

- **LPIPS vs PSIM:** Note that the Stable Diffusion (SD) VAE row reports PSIM metric, while other rows report LPIPS..
- **Reconstruction Accuracy:** Our custom VAE achieves a **high PSNR of 30.1** and **SSIM of 0.88**, preserving fine boundaries needed for text-to-image.

Three custom LDM variants scale from 40M to 450M parameters

LIGHT MODEL

~40M Parameters

- **Smallest Custom Variant:** Built to test the absolute lower bounds of text-to-image learning on COCO.
- **Inference Footprint:** Least computational complexity; operates extremely fast on minimal hardware.
- **Generation Quality:** Weakest visual reconstructions and text-alignment, acts as baseline proof-of-concept.



MEDIUM MODEL

~160M Parameters

- **Intermediate Scaling:** A mid-size custom U-Net designed to improve visual fidelity without excessive footprint.
- **Visual Performance:** Clearly outpaces the Light model in structure but still misses fine texture detail.
- **Balanced Baseline:** Serves as a great mid-point, but did not satisfy high-fidelity generation requirements.



STRONG MODEL

~450M Parameters

- **Largest Custom Model:** Main architecture used for the final comparative evaluation with CompVis LDM.
- **Memory Requirement:** 0.90 GB model weight and ~2.23 GB pipeline memory, highly feasible for edge devices.
- **Staged Training:** Optimized through initial training, checkpoint continuation, and Min-SNR loss fine-tuning.



Staged training and Min-SNR loss weighting optimize the Strong LDM

STAGED TRAINING PIPELINE (STRONG MODEL)

- **Stage 1: Scratch Training:** Initialized and trained the 450M Strong U-Net on cached 8-channel latents under standard v-prediction objective.
- **Stage 2: Resumed Continuation:** Continued training from the latest checkpoint under identical settings for **48 hours total on a single A100**.
- **Stage 3: Min-SNR Fine-tuning:** Resumed training with **Min-SNR loss weighting enabled**, shifting weight dynamically during final epochs.

THE POWER OF MIN-SNR LOSS WEIGHTING

- **The Challenge:** Diffusion training often struggles with noise imbalances across timesteps; high-SNR steps can dominate optimization.
- **The Solution: Min-SNR loss weighting** downweights very high-SNR timesteps to balance learning.
- **Timestep-Bucket Validation:** Focuses optimization efforts on late denoising steps which are crucial for detailed texture recovery.
- **The Outcome:** Balanced learning across the trajectory, yielding a noticeable drop in final validation FID.

DreamShaper depth pruning and distillation failed to recover coherence

THE EXPERIMENT

- **Pretrained Compression:** Attempted pruning pretrained Stable Diffusion models to recover quality faster than from-scratch training.
- **Target Model:** Selected **Lykon/dreamshaper-8** (SD1.5-style) after unsatisfactory trials on SD2/2.1.
- **Depth Reduction:** Cut intermediate blocks to create a compressed student model of **579M parameters** (originally ~859M).
- **Teacher-Student Distillation:** Student trained to match full DreamShaper teacher's denoising predictions on cached COCO latents.



THE RE-DISTILLATION FAILURE

- **Objective Improvements:** Distillation loss values steadily decreased over epochs, suggesting correct training convergence.
- **Visual Collapse:** Despite loss improvement, generated images were **completely unstable** and lacked spatial coherence.
- **Our explanation:** Depth-based pruning is **too aggressive** for latent U-Nets; critical structural priors are completely severed.
- **Decision:** Stopped experiment after several hours of limited compute. Training a compact model from scratch is far more robust.



Strong LDM outcompetes 872M reference model on FID with half the parameters

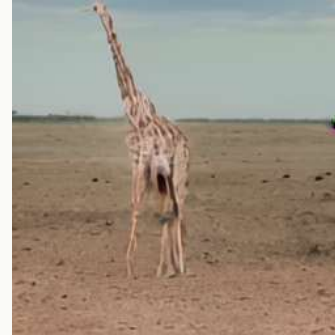
Model	Parameters	Best-FID Setting	FID ↓	KID ↓	IS ↑	CLIP ↑	Time / 32 imgs
Light	~40M	DDIM, 100 steps, CFG 5	135.13	0.0154	8.99	21.53	3.81 s
Medium	~160M	DDIM, 100 steps, CFG 5	117.35	0.0120	10.33	23.04	6.20 s
Strong (Ours)	~450M	DDIM, 100 steps, CFG 5	105.85	0.0076	12.92	24.45	8.59 s
LDM Reference	~872M	DDIM, 100 steps, CFG 5	113.94	0.0117	12.69	25.19	10.95 s

KEY METRIC INSIGHTS

- **The 450M Strong LDM Winner:** Our custom Strong LDM outcompetes the 872M CompVis reference model **on FID (105.85 vs 113.94)** and **KID (0.0076 vs 0.0117)**, proving highly targeted scratch training on specialized VAE space is superior.
- **Parameter Efficiency:** At half the parameter count, our Strong model achieves **higher Inception Score (12.92 vs 12.69)** and generates batches of 32 images **~21% faster (8.59 s vs 10.95 s)**.
- **The LDM Reference Advantage:** The CompVis 872M model still maintains stronger text-alignment (CLIP score: 25.19 vs 24.45) due to larger dataset pre-training weights.

VISUAL SYNTHESIS ANALYSIS

A lone giraffe walking in an open field.



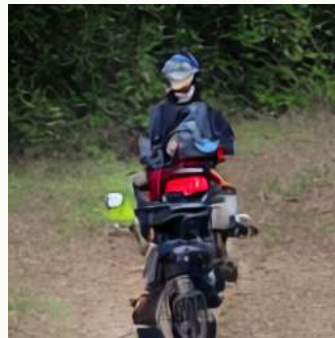
A bathroom sink sitting under a large mirror.



LDM
text2image



A man with a helmet on a small moped on a dirt road.



DDIM sampling accelerates inference 9.5x without quality degradation

Sampler	Steps	CFG	FID ↓	KID ↓	IS ↑	CLIP ↑	Avg Time / 32 imgs
DDIM	50	5	106.49	0.0086	12.42	24.57	4.54 s
DDIM (Best)	100	5	105.85	0.0076	12.92	24.45	8.59 s
DDIM	150	5	106.72	0.0082	12.41	24.68	12.66 s
DDPM (Baseline)	1000	5	106.97	0.0083	12.84	24.69	81.44 s

~9.5x SPEEDUP
DDIM 100 vs DDPM 1000

DDIM (100 steps) outcompetes DDPM (1000 steps) on FID (105.85 vs 106.97) while reducing generation time from **81.44 s to 8.59 s**.

~18x SPEEDUP
DDIM 50 vs DDPM 1000

DDIM (50 steps) yields an impressive FID of **106.49** in just **4.54 s**, preserving high image quality at minimal cost.

OPTIMAL PEAK
T=100 Optimal Trajectory

Denoising steps beyond 100 (T=150) slightly degrade FID (106.72), proving **100 steps** is the perfect convergence peak.

Low VRAM footprint enables deployment on edge devices

MEMORY PROFILE

3.17 GB

Peak Inference VRAM (fp16)

This released lightweight LDM checkpoint functions seems viably on edge devices. Operating inside restricted consumer GPUs, it eliminates massive system constraints.

AVERAGE INFERENCE BATCH TIMINGS AND MODEL WEIGHT (SECONDS / 32 IMAGES)

Model	Params	DDIM 100	DDIM 100	DDPM 1000	FP16 WEIGHT
Light	~40M	3.81 s	3.81 s	29.74 s	0.08 GB
Medium	~160M	6.20 s	6.20 s	55.57 s	0.32 GB
Strong	~450M	8.59 s	8.59 s	81.44 s	0.90 GB
Reference LDM	~872M	10.95 s	10.95 s	103.06 s	1.74 GB

- **The Parallel GPU Paradox:** Our 450M Strong model compiles and runs **faster** than the Reference LDM (**8.59 s vs 10.95 s** at 100 steps).
- **Extreme Edge Viability:** Switch to DDIM with 50 steps; our Strong model yields a batch in **4.54 seconds**, enabling interactive local testing.
- **FP16 Weight Footprint:** The Strong checkpoint requires ~0.90 GB for fp16 weights, compared with **1.74 GB** for the Reference LDM, making it easier to store, load, and deploy on constrained GPUs.

Future Work:

STEP 1: HIGHER SPATIAL RESOLUTION

Current model generates 256×256 images through 32×32 latent tensors.

A natural next step is moving to 512×512 generation, either by training on

higher-resolution images or by adding a lightweight super-resolution stage.

This would improve fine details, object boundaries, and text-image realism,

but requires more memory and longer training.

STEP 2: LARGER TRAINING SCALE

The model was trained under limited compute, so scaling the dataset size, training duration, and caption diversity would likely improve semantic alignment and reduce artifacts.

More diverse data would especially help object composition, rare concepts, and prompt-following quality.

STEP 3: FURTHER MODEL COMPRESSION

Future work can combine pruning, quantization, and knowledge distillation to reduce the U-Net and text encoder footprint further.

The goal would be to keep the DDIM speed advantage while making the model deployable on smaller consumer or edge GPUs.

Thank you for your attention!

Happy to take questions.

Project: Lightweight Latent Diffusion Models

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